Assignment 12

FTP & SNMP

1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Requirements:

- Windows 11
- FileZilla Server software
- Internet connection

Procedure:

Step 1: Download and Install

1. I opened Google Chrome and searched for FileZilla Server.
2. I visited the official FileZilla website and downloaded FileZilla Server.
3. I opened the installer and completed the installation.
4. I launched FileZilla Server.

Step 2: Create a User Account

1. I opened the FileZilla Server administration interface.
2. I navigated to the Users section.
3. I created a new user account with a username and password.
4. I saved the user settings.

Step 3: Configure a Shared Folder

1. I selected the newly created user.
2. I opened the Mount Points / Shared Folders settings.
3. I added a folder from my computer.
4. I assigned read and write permissions to the folder.
5. I saved the configuration.

Step 4: Verify FTP Access

I checked the FTP server settings and verified that the user could access the shared folder.

Result: Successfully installed FileZilla Server, created a user account, and configured a shared folder for FTP access.

2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Procedure:

Step 1: Connect to FTP Server

1. I opened FileZilla Client on my Windows computer.
2. I entered the FTP server IP address (127.0.0.1), username, password, and port (21).
3. I clicked Quickconnect to connect to the FTP server.

Step 2: Upload an Image

1. After connecting, I located an image file on my computer in the left panel.
2. I opened the shared folder in the right panel.
3. I selected the image file and dragged it from the left panel to the right panel.
4. I waited for the upload to complete.

Step 3: Verify the Upload

I checked the right panel to confirm that the uploaded image file was available in the FTP shared folder.

Result: Successfully connected to the FTP server using FileZilla Client and uploaded an image file to the shared folder.

3. **Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.**

Procedure:

Step 1: Connect to FTP Server

1. I opened FileZilla Client.
2. I entered the server IP address (127.0.0.1), username, password, and port (21).
3. I clicked Quickconnect to connect to the FTP server.

Step 2: Download the File

1. I located the previously uploaded image in the right panel (server).
2. I selected a different folder on my computer in the left panel.
3. I right-clicked the image file and selected Download.
4. I waited for the download to complete.

Step 3: Verify File Size

1. I opened the original image folder on my computer.
2. I checked the original file size using Properties.
3. I checked the downloaded file size in the same way.
4. I compared both file sizes to confirm they matched.

Result: Successfully downloaded the image file from the FTP server to a different folder and verified that its file size matched the original file.

4. **Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites). Hint: Think about how network admins monitor device health, bandwidth, or uptime.**

1. Internet Service Providers (ISPs)

- Monitor routers and network traffic.
- Check bandwidth usage and connectivity.
- Detect network failures.
- Example: Jio and Airtel.

2. College Campus WiFi

- Monitor WiFi routers and access points.
- Check connected devices and bandwidth.
- Identify network performance issues.

3. Large Websites and Data Centers

- Monitor servers, routers, and switches.
- Check network uptime and device health.
- Detect hardware and network failures.
- Example: Google and Amazon.

5. **Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.**

Points	**FTP**	**HTTP**
Full Form	File Transfer Protocol	Hypertext Transfer Protocol
Port Number	21 (Control), 20 (Data)	80
Security	Not encrypted by default	Not encrypted by default
Data Transfer	Transfers files between computers	Transfers web pages and web content
Use Cases	Uploading and downloading files	Browsing websites
Authentication	Supports username and password	Usually does not require authentication
Secure Version	FTPS / SFTP	HTTPS (Port 443)